



PHASE 03 • FORTIFY

Build the platform . Prove it works.
Defend the system you designed, live in production.

duothan.ieeeensbm.org

01 • CONTEXT

From Reality to Resilience

A massive global cyber disaster in the year 2065, caused by the Super Malware Agent, has taken down systems across governments, businesses, healthcare, transportation, and financial services. The entire world has been brought to a standstill.

In the financial sector, the damage is severe. Core banking systems, ATMs, digital payment platforms, and loan systems have completely stopped working. Millions of people have lost access to essential financial services. Small businesses can no longer process digital transactions, and people are forced to rely entirely on cash. This loss of secure banking has slowed economic recovery and increased financial inequality across the world.

The Master Key has been recovered and the banking platform has been rebuilt. Customer data is safe, the services are written, and the architecture holds. But a platform that only runs inside a repository saves no one. Millions of people are still locked out of their money, and the world is waiting for the banks to come back online.

The mission of Duothan 6.0 is to save the digital banking system

Deploying that platform to the cloud, releasing it through automation, and operating it as a service the world can depend on, running as **independent services**.

A working application is not a restored financial system. The platform must survive real traffic, real failures, and real attackers. Every release must be repeatable, every environment reproducible, every secret protected, and every incident visible long before a customer feels it.

Your mission is to take the system out of the repository and into production, and to prove it can stay there.

02 • BRIEF**Phase Overview**

In Phase 3 - Restore, teams move from a working application to a running production system. This phase focuses on deploying the banking platform built during Phase 2 onto a cloud environment as a set of independent services, and on building the automation around it that makes every release repeatable, verifiable, and safe. The deployed solution should remain consistent with the architecture proposed in Phase 1 and the functionality delivered in Phase 2, now operating as a live and publicly accessible service.

Teams are expected to move their solution out of the development environment and into a live, reliable service. The platform should behave the same way wherever it runs, recover from failure without manual intervention, and remain available as demand changes. Teams should be able to release an update confidently and repeatedly, and to understand the state of the running system at any moment rather than discovering problems through their users.

Security and engineering discipline are expected throughout the delivery process rather than at the end of it. Every change should reach production through the same controlled and repeatable route, with risks identified before release and sensitive information protected at every stage. The detailed requirements for this phase are set out in the Technical Scope that follows. Teams are free to decide how they meet them, provided their approach follows industry best practices in the industry and can be justified during evaluation.

03 · SUBMISSION

Final Deliverables

At the end of Phase 3, your team submits the following:

01	Public GitHub Repository Link
02	Live Deployed Application URL or IP Address
03	Brief Deployment Documentation with Screenshots

SUBMISSION FORMAT

Teams must submit the **Public GitHub Repository Link** and the **Live Deployed Application URL or IP Address of the application** through the submission form. Ensure the repository is public and contains the application code, infrastructure definitions, and pipeline configuration, and that the deployed application is reachable and functional before submission.

Submission Link : duothan.ieeensbm.org/submission

04 · EVALUATION

Mark Allocation

CRITERIA	WEIGHT
Service Deployment & Environment Consistency	15%
Build & Release Automation	20%
Automated Infrastructure & Configuration Management	15%
Scalability, Availability & Reliability	10%
Operational Visibility & System Health	15%
Security Practices & Protection of Sensitive Data	15%
Engineering Best Practices	5%
Team contributions	5%
TOTAL	100%

Note: the above weightings are indicative and may be adjusted by the judging panel to reflect the depth of execution shown across sections.

BY THE END OF PHASE 3

Your team should have a digital banking platform running in production, released and operated entirely through automation.

The platform rebuilt during REBUILD goes live during RESTORE.

The world is waiting for its banks to come back online.

Deploy it. Automate it. Keep it running.

Duothan 6.0 · IEEE Student Branch of NSBM

Questions? duothan.ieeensbm.org